EE6380 Deep Learning

Exercise problems

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These are exercise problems for you to work out. You do not have to submit the solutions

Problem 1 (Uniform convergence $\Rightarrow$ ERM is an agnostic PAC learner)**.** Let $\mathcal{H}$ be a finite hypothesis class and let the loss be bounded, $\ell \in [0, 1]$.

(a) Fix $h \in \mathcal{H}$. Using Hoeffding's inequality, show that for every $\epsilon' > 0$,

$$\mathbb{P}\big[\, |R(h) - R_{\mathcal{S}}(h)| > \epsilon' \,\big] \leq 2e^{-2n(\epsilon')^2}.$$

(b) Deduce the *uniform convergence* property: with probability at least $1-\delta'$, $|R(h)-R_{\mathcal{S}}(h)| \leq \epsilon'$ *for all* $h \in \mathcal{H}$, provided $n \geq \frac{1}{2(\epsilon')^2} \log \frac{2|\mathcal{H}|}{\delta'}$.

(c) Let $h_{\mathrm{erm}} = \arg\min_{h\in\mathcal{H}} R_{\mathcal{S}}(h)$. Show that on the above event, $R(h_{\mathrm{erm}}) \leq R^*(\mathcal{H}) + 2\epsilon'$. Conclude that $\mathcal{H}$ is agnostic PAC learnable with sample complexity

$$N(\epsilon, \delta) = \left\lceil \tfrac{2}{\epsilon^2} \log \tfrac{2|\mathcal{H}|}{\delta} \right\rceil.$$

Problem 2 (Computing VC dimensions)**.** Determine the VC dimension of each class (with proofs of both the lower and upper bounds).

(a) Axis-aligned rectangles in the plane: $\mathcal{H} = \{\mathbf{1}\{\mathbf{x} \in [a_1, b_1] \times [a_2, b_2]\}\}$ on $\mathcal{X} = \mathbb{R}^2$.

(b) Affine halfspaces in $\mathbb{R}^d$: $\mathcal{H} = \{\mathbf{1}\{\mathbf{w}^\top\mathbf{x} + b \geq 0\} : \mathbf{w} \in \mathbb{R}^d,\, b \in \mathbb{R}\}$.

(c) Any finite class: show $\mathrm{VCdim}(\mathcal{H}) \leq \log_2 |\mathcal{H}|$.

Problem 3 (VC dimension $\neq$ number of parameters: $\sin(ax)$)**.** Let $\mathcal{X} = \mathbb{R}$, $\mathcal{Y} = \{0, 1\}$ and $\mathcal{H} = \{h_a(x) = \mathbf{1}\{\sin(ax) \geq 0\} : a \in \mathbb{R}\}$, a one-parameter family. Prove $\mathrm{VCdim}(\mathcal{H}) = \infty$ by showing that for every m the points $x_i = 2^{-i}$, $i = 1, \ldots, m$, are shattered.

Problem 4 (Which sets are convex?)**.** For each set, state whether it is convex and give a rigorous justification.

(a) Halfspace $\{\mathbf{x} : \mathbf{a}^\top\mathbf{x} \leq b\}$.

(b) Hyperplane $\{\mathbf{x} : \mathbf{a}^\top\mathbf{x} = b\}$.

(c) Euclidean ball $\{\mathbf{x} : \|\mathbf{x} - \mathbf{x}_0\|_2 \leq r\}$.

(d) Annulus $\{\mathbf{x} : 1 \leq \|\mathbf{x}\|_2 \leq 2\}$.

(e) Box $\{\mathbf{x} : \ell_i \leq x_i \leq u_i,\ i = 1, \ldots, d\}$.

(f) The two-point set $\{\mathbf{0}, \mathbf{v}\}$ with $\mathbf{v} \neq \mathbf{0}$.

(g) Complement of a ball $\{\mathbf{x} : \|\mathbf{x}\|_2 \geq 1\}$.

(h) Probability simplex $\Delta = \{\mathbf{x} : x_i \geq 0,\ \sum_i x_i = 1\}$.

Problem 5 (Intersections vs. unions)**.** (a) Prove that the intersection $\bigcap_\alpha \mathcal{A}_\alpha$ of any family of convex sets is convex.

(b) Show by example that the union of two convex sets need not be convex. Under what simple condition *is* $\mathcal{A}_1 \cup \mathcal{A}_2$ convex?

Problem 6 (Affine images and minimizer sets). (a) Let $\mathcal{C} \subseteq \mathbb{R}^d$ be convex and $g(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ an affine map. Prove the image $g(\mathcal{C}) = \{A\mathbf{x} + \mathbf{b} : \mathbf{x} \in \mathcal{C}\}$ is convex.

(b) Let f be convex. Prove its set of global minimizers $\arg\min_{\mathbf{x}} f(\mathbf{x})$ is convex.

Problem 7 (Second-derivative classification). For each function, use f'' to classify it as convex, concave, or neither on the stated domain.

(a) $f(x) = x^2$ on $\mathbb{R}$.
(b) $f(x) = x^3$ on $\mathbb{R}$.
(c) $f(x) = e^{ax}$ on $\mathbb{R}$ (any $a \in \mathbb{R}$).
(d) $f(x) = \log x$ on $(0, \infty)$.
(e) $f(x) = x \log x$ on $(0, \infty)$.
(f) $f(x) = 1/x$ on $(0, \infty)$.
(g) $f(x) = \sqrt{x}$ on $(0, \infty)$.

Problem 8 (Affine functions are the only both-convex-and-concave functions). (a) Show every affine $f(x) = ax + b$ is both convex and concave.

(b) Show conversely: if a (twice-differentiable) f is both convex and concave, then it is affine.

Problem 9 (ReLU and absolute value). (a) Show the ReLU $\phi(x) = \max\{0, x\}$ is convex.

(b) Show $f(x) = |x|$ is convex but *not* strongly convex, and give its subdifferential at $x = 0$.

Problem 10 (Strong convexity and smoothness constants of x^2). Let $f(x) = x^2$.

(a) Find the largest μ for which f is μ-strongly convex.
(b) Find the smallest L for which f is L-smooth (f' is L-Lipschitz).
(c) What is the condition number $\kappa = L/\mu$, and what does it say about GD?

Problem 11 (Quadratic forms). Let $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top Q\mathbf{x} + \mathbf{b}^\top\mathbf{x} + c$ with $Q = Q^\top \in \mathbb{R}^{d\times d}$.

(a) Compute $\nabla f(\mathbf{x})$ and $\nabla^2 f(\mathbf{x})$.
(b) Show f is convex iff $Q \succeq 0$, and strictly convex iff $Q \succ 0$.
(c) For $d = 2$, $Q = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 8 \end{smallmatrix}\right)$: is f convex? strongly convex? Give μ, L and κ.

Problem 12 (Epigraph characterization). The epigraph of $f : \mathbb{R}^d \to \mathbb{R}$ is $\mathrm{epi}(f) = \{(\mathbf{x}, t) \in \mathbb{R}^{d+1} : t \geq f(\mathbf{x})\}$. Prove that f is convex if and only if $\mathrm{epi}(f)$ is a convex set.

Problem 13 (Sublevel sets: a one-way street). The α-sublevel set of f is $S_\alpha = \{\mathbf{x} : f(\mathbf{x}) \leq \alpha\}$.

(a) Prove that if f is convex then every S_α is convex.
(b) Show the converse is false: give a function whose sublevel sets are all convex but which is not itself convex.

Problem 14 (Convexity via restriction to lines). Prove that $f : \mathbb{R}^d \to \mathbb{R}$ is convex if and only if, for every $\mathbf{x} \in \mathbb{R}^d$ and every direction $\mathbf{v} \in \mathbb{R}^d$, the one-dimensional function $g_{\mathbf{x},\mathbf{v}}(t) = f(\mathbf{x} + t\mathbf{v})$ is convex in $t \in \mathbb{R}$.

Problem 15 (Pointwise maximum and supremum of affine functions). (a) Show that if f_1, f_2 are convex then $f(\mathbf{x}) = \max\{f_1(\mathbf{x}), f_2(\mathbf{x})\}$ is convex.

(b) More generally, show that a supremum of affine functions, $f(\mathbf{x}) = \sup_{i\in I}(\mathbf{a}_i^\top\mathbf{x} + b_i)$, is convex (for *any* index set I).

Problem 16 (Convexity zoo). For each function decide convex / concave / neither, and give a (sub)gradient where one exists. Argue rigorously.

(a) Hinge loss $u \mapsto \max\{0, 1-u\}$ (here $u = yy'$).
(b) Logistic loss $u \mapsto \log(1+e^{-u})$.
(c) Log-sum-exp $f(\mathbf{w}) = \log \sum_{i=1}^{d} e^{w_i}$.
(d) $f(\mathbf{w}) = \max\{w_1, \ldots, w_d\}$.
(e) $f(\mathbf{w}) = \|\mathbf{w}\|$ for an arbitrary norm.

Problem 17 (Operations that preserve convexity)**.** Prove the following (all functions real-valued).
(a) Nonnegative combinations: if f_1, f_2 are convex and $\lambda_1, \lambda_2 \geq 0$, then $\lambda_1 f_1 + \lambda_2 f_2$ is convex.
(b) Affine precomposition: if f is convex then $g(\mathbf{w}) = f(A\mathbf{w} + \mathbf{b})$ is convex.
(c) Separable sums: if each $f_i : \mathbb{R} \to \mathbb{R}$ is convex then $f(\mathbf{x}) = \sum_i f_i(x_i)$ is convex.
(d) Scalar composition: if f_2 is convex and f_1 is convex and *nondecreasing*, then $f_1 \circ f_2$ is convex. Give a counterexample showing convexity of both is not sufficient in general.

Problem 18 (Quadratics, conditioning, and eigenvalues)**.** Let $A \in \mathbb{R}^{d\times d}$ and $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$.
(a) Show we may assume $A = A^\top$ WLOG, and compute $\nabla f, \nabla^2 f$.
(b) For which A is f convex? μ-strongly convex?
(c) If f is μ-strongly convex and L-smooth, express μ, L and the GD condition number κ in terms of the eigenvalues of A.

Problem 19 (L_2-regularized logistic regression: convexity, strong convexity, smoothness)**.** Data $(\mathbf{x}_i, y_i)$, $y_i \in \{0,1\}$, $\sigma(u) = 1/(1+e^{-u})$, and per-example loss $\ell_i(\mathbf{w}) = -y_i \ln \sigma(\mathbf{w}^\top \mathbf{x}_i) - (1 - y_i)\ln(1-\sigma(\mathbf{w}^\top \mathbf{x}_i))$. Consider

$$F(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n} \ell_i(\mathbf{w}) + \frac{\lambda}{2}\|\mathbf{w}\|_2^2 .$$

Compute ∇F and $\nabla^2 F$, show F is convex, and determine the best strong-convexity and smoothness constants μ, L. (Let $X \in \mathbb{R}^{n\times d}$ have rows $\mathbf{x}_i^\top$.)

Problem 20 (SGD: unbiasedness, variance, and the noise floor)**.** Let $f(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n} f_i(\mathbf{w})$.
(a) For SGD, $I \sim \mathrm{Unif}[n]$ and $F'(\mathbf{w}) = \nabla f_I(\mathbf{w})$. Show $\mathbb{E}[F'(\mathbf{w})] = \nabla f(\mathbf{w})$.
(b) For minibatch SGD with $\mathcal{I}$ a size-B sample *with replacement* and $F'(\mathbf{w}) = \frac{1}{B}\sum_{i\in\mathcal{I}} \nabla f_i(\mathbf{w})$, show it is unbiased with $\mathbb{E}\|F'(\mathbf{w}) - \nabla f(\mathbf{w})\|^2 = \sigma^2(\mathbf{w})/B$, where $\sigma^2(\mathbf{w}) = \frac{1}{n}\sum_i \|\nabla f_i(\mathbf{w}) - \nabla f(\mathbf{w})\|^2$.
(c) For μ-strongly convex L-smooth f, constant step $\rho \leq 1/L$, and bounded-variance stochastic gradients ($\mathbb{E}\|F'(\mathbf{w}) - \nabla f(\mathbf{w})\|^2 \leq \sigma^2$), prove the one-step contraction $\mathbb{E}\|\mathbf{w}^+ - \mathbf{w}^\star\|^2 \leq (1 - \rho\mu)\mathbb{E}\|\mathbf{w} - \mathbf{w}^\star\|^2 + \rho^2\sigma^2$ and identify the resulting asymptotic "noise floor."

Problem 21 (Subgradient descent need not decrease the objective)**.** Let $f(w_1, w_2) = |w_1| + 2|w_2|$ (convex, nonsmooth). At $\mathbf{w} = (1,0)$ take the subgradient $\mathbf{g} = (1,2) \in \partial f(\mathbf{w})$ (valid since $2 \in \partial|w_2|$ at 0). For step size η, evaluate $f(\mathbf{w} - \eta\mathbf{g})$ and show the objective can strictly *increase*. Explain why subgradient methods output $\arg\min_{1\leq t\leq T} f(\mathbf{w}^{(t)})$.

Problem 22 (Backprop through a skip connection (fan-out))**.** Consider $\mathbf{z} = \phi(W_1\mathbf{x})$, $\hat{\mathbf{y}} = W_2\mathbf{z} + W_3\mathbf{x}$ (a linear skip), squared loss $L = \frac{1}{2}\|\hat{\mathbf{y}} - \mathbf{y}\|^2$, with ϕ applied elementwise. Derive $\partial L/\partial W_1, \partial L/\partial W_2, \partial L/\partial W_3$ and $\partial L/\partial \mathbf{x}$. Point out where fan-out (summation of adjoints) occurs.

Problem 23 (Weight sharing: fan-out over time (BPTT))**.** An RNN shares one matrix W across steps: $\mathbf{z}_0 = \mathbf{x}$, $\mathbf{z}_t = \phi(W\mathbf{z}_{t-1})$ for $t = 1, 2$, and $L = \ell(\mathbf{z}_2)$. Because the *same* W is used twice, show

$$\frac{\partial L}{\partial W} = \boldsymbol{\delta}_2\, \mathbf{z}_1^\top + \boldsymbol{\delta}_1\, \mathbf{z}_0^\top, \quad \text{with } \boldsymbol{\delta}_2 = \phi'(W\mathbf{z}_1) \odot \nabla_{\mathbf{z}_2}\ell, \quad \boldsymbol{\delta}_1 = \phi'(W\mathbf{z}_0) \odot (W^\top \boldsymbol{\delta}_2).$$

Explain the general rule for shared parameters.

Problem 24 (Gradient checkpointing: the memory–compute tradeoff)**.** A depth-m network stores activations for the backward pass.

(a) What are the memory and compute costs of vanilla reverse mode (store everything)?

(b) Suppose we keep only k evenly spaced checkpoints and recompute intermediate activations within a segment during the backward pass. Give the memory cost and the extra compute cost as functions of m, k.

(c) Minimize total activation memory $\approx k + m/k$ over k and report the optimal k and the resulting memory.

Problem 25 (LDA yields the logistic posterior)**.** Two classes with priors $\pi_1 = P(Y = 1), \pi_0 = P(Y = 0)$ and Gaussian class-conditionals $P_{X|Y=y} = \mathcal{N}(\mu_y, \Sigma)$ sharing covariance Σ. Show the posterior is exactly logistic,

$$P(Y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b), \qquad \mathbf{w} = \Sigma^{-1}(\mu_1 - \mu_0),$$

and find b. What changes if $\Sigma_1 \neq \Sigma_0$?

Problem 26 (Softmax cross-entropy: gradient, convexity, smoothness)**.** Logits $\mathbf{z} = W\mathbf{x} \in \mathbb{R}^k$, $\mathbf{p} = \mathrm{softmax}(\mathbf{z})$, one-hot label $\mathbf{y}$, and $L = -\sum_c y_c \log p_c$.

(a) Show $\nabla_{\mathbf{z}} L = \mathbf{p} - \mathbf{y}$ and $\nabla^2_{\mathbf{z}} L = \mathrm{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$.

(b) Show L is convex in $\mathbf{z}$ and 1-smooth (in fact $\frac{1}{2}$-smooth); show it is *not* strongly convex.

(c) Conclude convexity/smoothness in W, and comment on strong convexity.

Problem 27 (Metrics, class imbalance, and the ROC curve)**.** (a) For the screening table $N_{TP} = 40$, $N_{FP} = 100$, $N_{FN} = 10$, $N_{TN} = 850$ (50 diseased of 1000), compute accuracy, recall, precision, FPR, FDR, and F_1. Comment on why accuracy misleads.

(b) As the decision threshold τ on a score $\beta(\mathbf{x})$ decreases from $+\infty$ to $-\infty$, show that both TPR and FPR are nondecreasing, so the ROC curve is monotone from $(0, 0)$ to $(1, 1)$.

(c) State and briefly justify the probabilistic meaning of the area under the ROC curve (AUC).

Problem 28 (Probit as a latent-variable model)**.** Define a latent score $Z = \mathbf{w}^\top \mathbf{x} + \varepsilon$ and set $Y = 1$ iff $Z \geq 0$.

(a) If $\varepsilon \sim \mathcal{N}(0, 1)$, show $P(Y = 1 \mid \mathbf{x}) = \Phi(\mathbf{w}^\top \mathbf{x})$ (probit).

(b) If ε has the standard logistic distribution (CDF $\sigma(t) = 1/(1 + e^{-t})$), show $P(Y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x})$ (logistic regression).

(c) Why is the negative log-likelihood convex in $\mathbf{w}$ in both cases?